



Ceylon Rail Intelligence

AI-Driven Intelligent Railway Operations System for Sri Lanka

Project ID: R26-IT-127

Supervisor: Ms. Chathurangika

Kahadawaarachchi

Team Members



IT22217240 – Narasinghe N.K.B.N.N



IT22051202 – Didulantha C.S

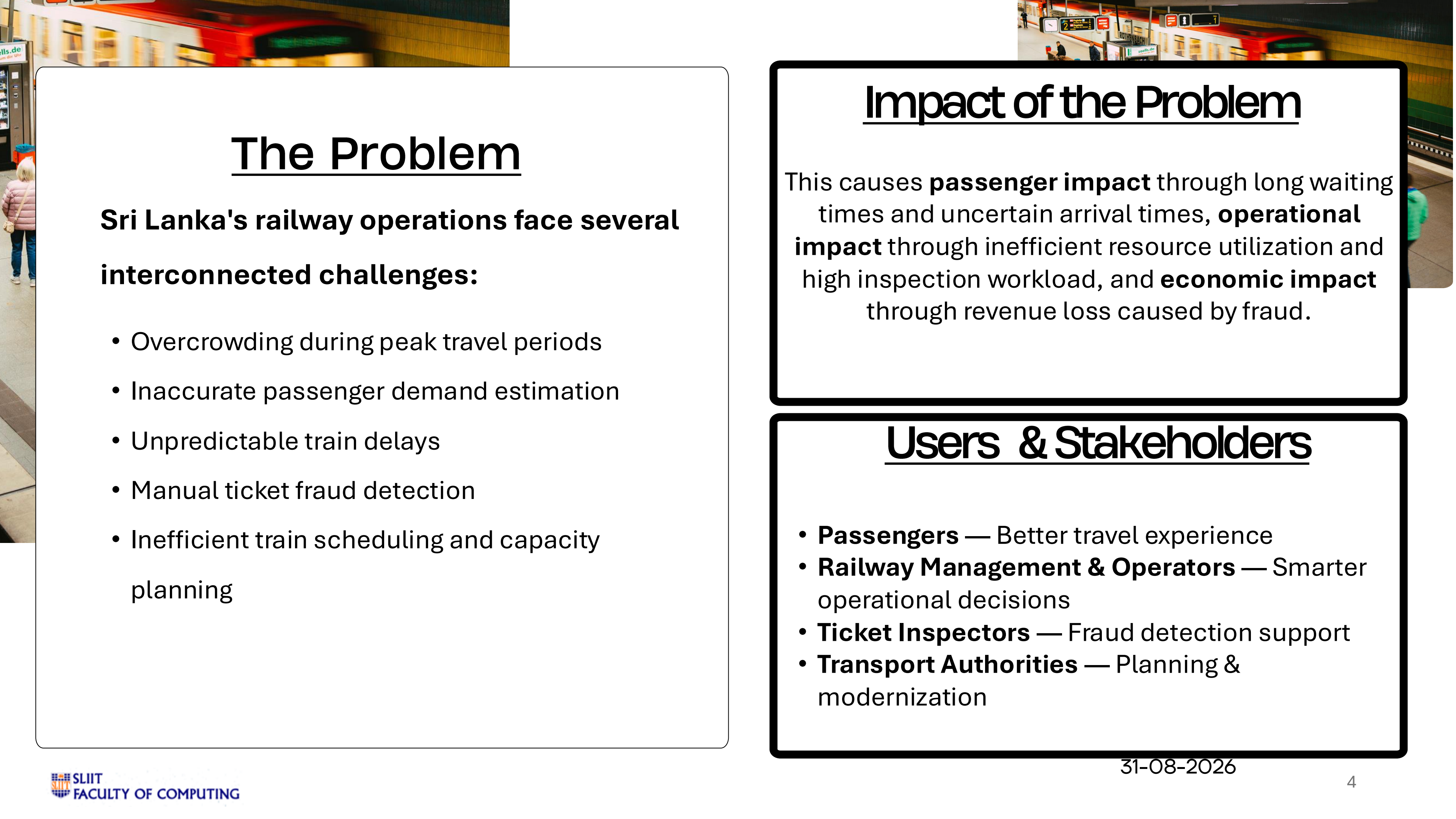


IT22126610 – Muthukudaarachchi V.U



IT22151506 – D.M.N.T Senawirathna





The Problem

Sri Lanka's railway operations face several interconnected challenges:

- Overcrowding during peak travel periods
- Inaccurate passenger demand estimation
- Unpredictable train delays
- Manual ticket fraud detection
- Inefficient train scheduling and capacity planning

Impact of the Problem

This causes **passenger impact** through long waiting times and uncertain arrival times, **operational impact** through inefficient resource utilization and high inspection workload, and **economic impact** through revenue loss caused by fraud.

Users & Stakeholders

- **Passengers** — Better travel experience
- **Railway Management & Operators** — Smarter operational decisions
- **Ticket Inspectors** — Fraud detection support
- **Transport Authorities** — Planning & modernization

Research Gap

- Existing systems lack AI-driven railway analytics
- Limited real-time forecasting capabilities
- No adaptive seat allocation based on predicted demand
- Lack of integrated intelligent railway decision-support systems

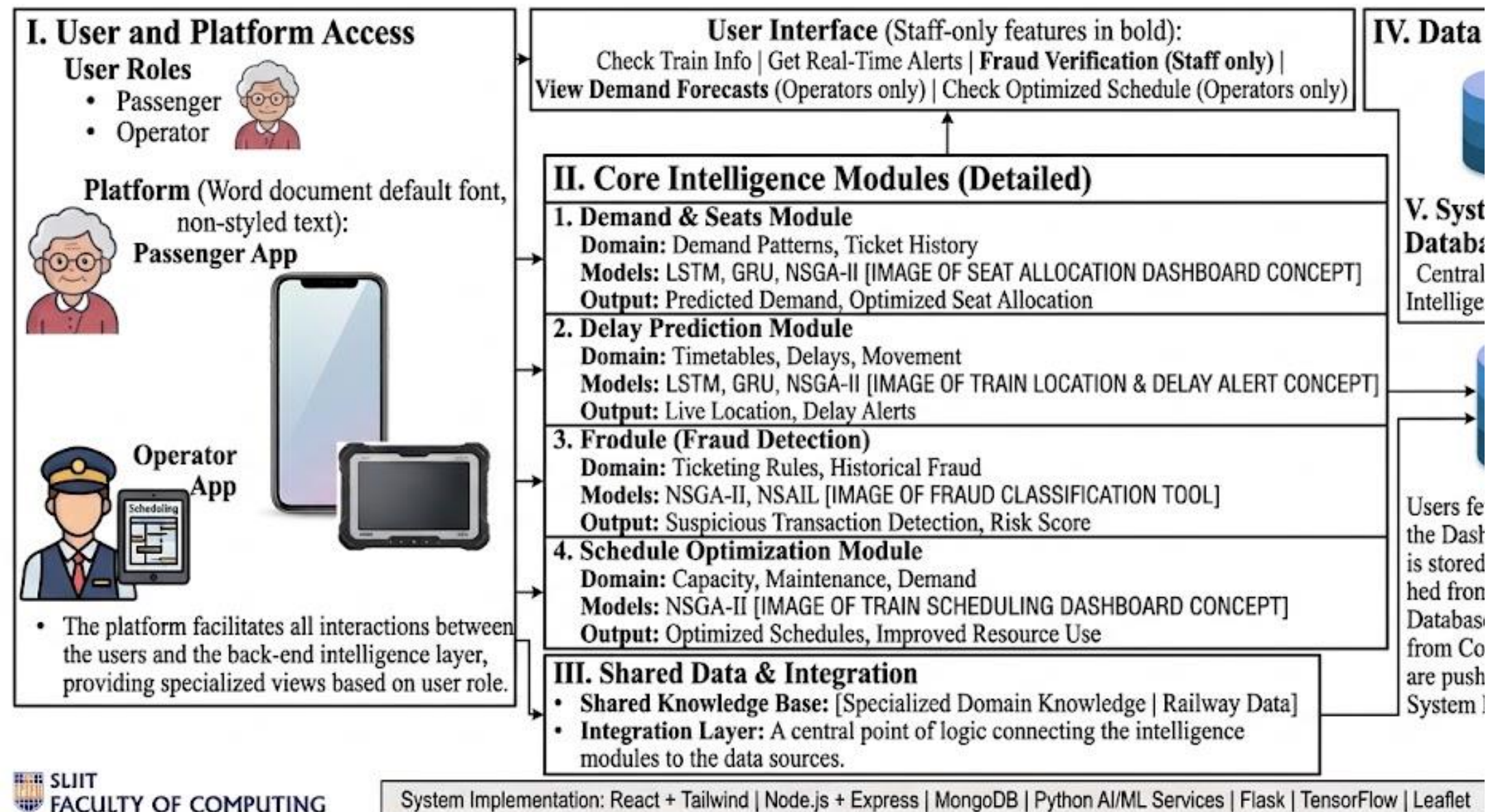
Key Objectives

- Forecast passenger demand and optimize seat allocation.
- Predict train delays using predictive models.
- Detect suspicious ticket transactions using AI.
- Optimize train schedules considering multiple operational objectives.

Research Gap & Key Objectives

Solution Overview

Our solution integrates four intelligent components to address key operational challenges in Sri Lanka's railway system:



1. Passenger Demand Forecasting & Adaptive Seat Allocation

Uses AI to predict passenger demand across different routes and time periods.

Based on predicted demand, the system supports adaptive seat allocation and capacity planning .

2. Real-Time Train Tracking & Delay Prediction

Uses train location and timetable data to provide real-time train tracking and predict potential delays.

This helps passengers and railway operators make better travel and operational decisions.

3. AI-Based Railway Fraud Detection & Passenger Verification

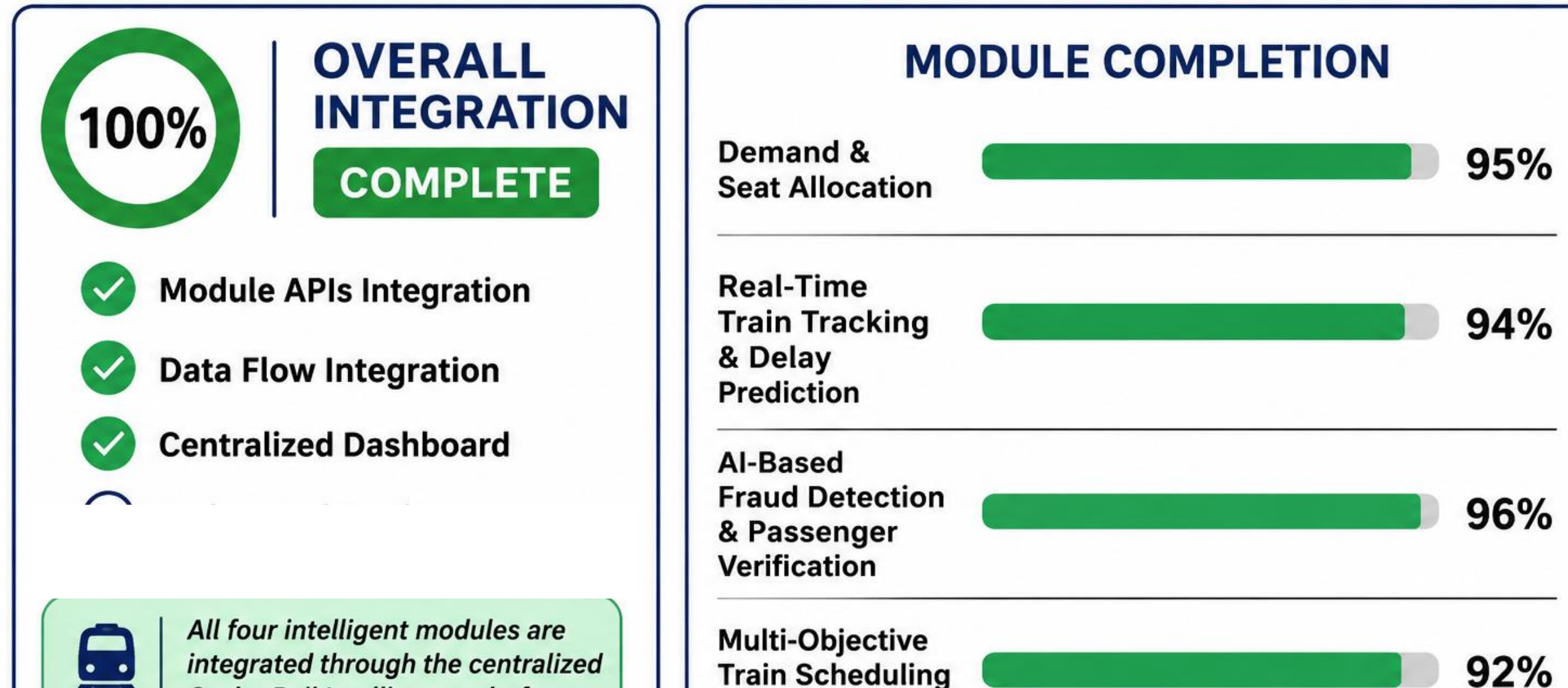
Analyzes ticket transactions using machine learning to identify suspicious activities and classify fraud risk.

The system supports railway operators with real-time alerts.

4. Multi-Objective Train Scheduling Optimization

Uses the NSGA-II optimization framework to generate improved train schedules while considering multiple factors such as passenger demand, train capacity, congestion, delays, and resource utilization.

Integration And Completion Status



Remaining Work

- Final end-to-end testing
- Performance optimization
- Model refinement
- Integration validation
- UI/UX improvements
- Deployment preparation

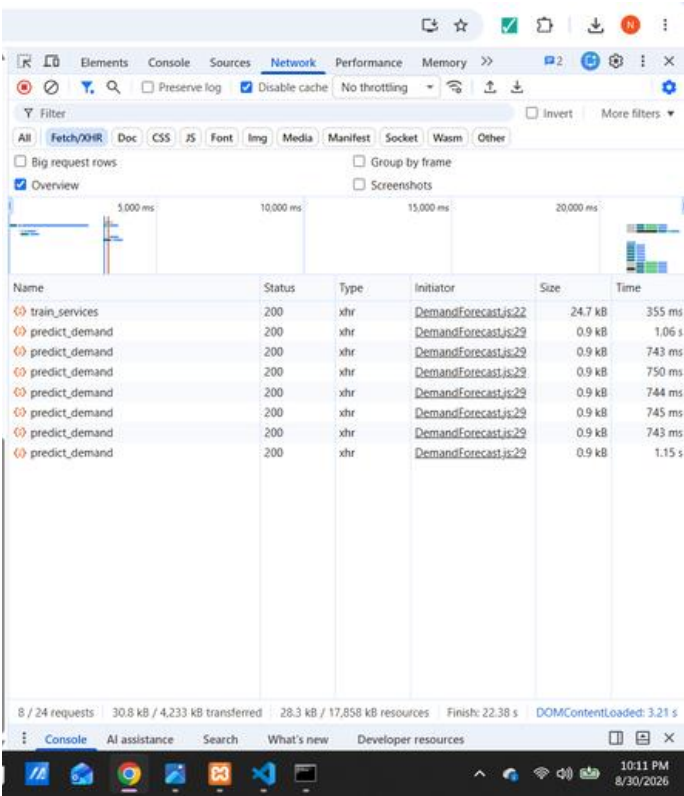
All individual research modules have reached over 90% implementation completion and are integrated into the overall CeylonRail Intelligence system.



Technical Evaluation Evidence

Class-wise Demand Forecasting Evaluation

```
class_wise_metrics.csv
backend > public > outputs > class_wise_metrics.csv
1 class,MAE,RMSE,MAPE_%
2 1st Class,3.6,4.51,13.11
3 2nd Class,21.45,26.82,13.09
4 3rd Class,47.27,59.32,13.15
5
```



Non-Functional Requirements

NFR	Evidence / How Demonstrated
Performance	Response time and system testing
Scalability	Modular architecture and API-based services
Security	Authentication and role-based access
Reliability	Error handling and validation
Usability	User-friendly web/mobile interfaces
Maintainability	Independent modules and modular architecture

Data Collection & Stakeholder Evidence

In the last PP1, as instructed by the p... We collected data collection and gaining domain knowledge by meeting relevant railway stakeholders and gathering real-world insights. We conducted data collection and domain validation with relevant railway stakeholders to better understand the existing processes, challenges, and requirements.



Historical Ticketing & Passenger Data from PCO unit (SLR) collected.
Date 12/08/2026.
[Signature]



K.H.M.M. Wijekoon
Assistant Director (Costing)
Principal Costing Officer Of
Sri Lanka Railway Department
Colombo 10.

tickets price vs. season tickets price rates, class-based yield management, so the forecasting model reflects real-world revenue optimization.

- **Operational Constraints Alignment:** Ensure the adaptive seat allocation algorithm factors in practical constraints such as fixed rake/carriage capacities, locomotive power limitations, and turnaround times at major junctions.
- **Data Sources & Anomaly Handling:** Test how the forecasting model accounts for sudden demand spikes caused by unexpected weather events, holiday travel, or sudden service disruptions.
- **Impact Metrics:** Quantify the financial and operational benefits (e.g., projected decrease in unreserved congestion, percentage increase in revenue yield per kilometer).

Overall, this is a promising initiative with significant potential. Keep refining the model based on operational realities, and you will have a very strong presentation for your final evaluation.

Best regards,
Menaka Wijekoon.
Assistant Director (Costing),
Sri Lanka Railways.

Commercialization Roadmap: Pilot to National Scale



Start Small. Prove Value. Scale Nationally.

**Pilot Railway Operations → Department Deployment → National
Transport Intelligence**



Thankyou!